Rebuilding the Analog World: DAC Fundamentals

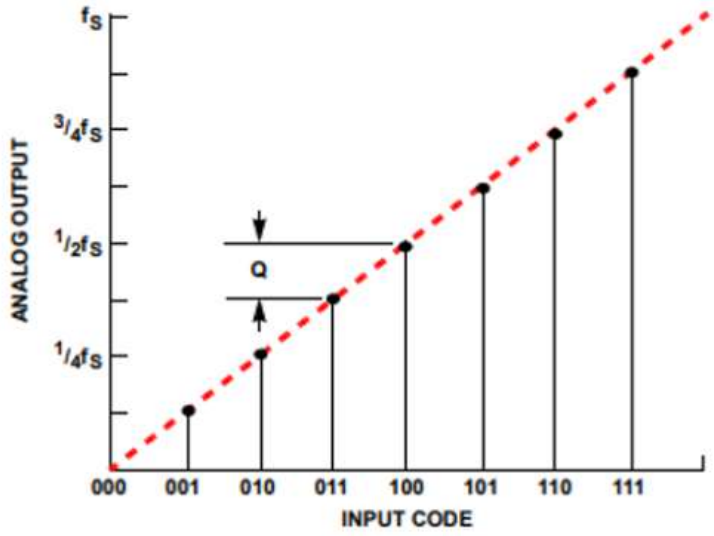
The Inverse Operation: A DAC converts a digital code (N-bit binary number) into a proportional analog output (typically voltage or current).

Click the buttons to learn more.

Ideal Transfer Function:

- A perfect staircase where each step corresponds to a 1 LSB change in the digital input code.
- The analog output is given by:

$$V_{OUT} = V_{REF} \cdot \left(\frac{\text{Digital Code}}{2^N}\right)$$

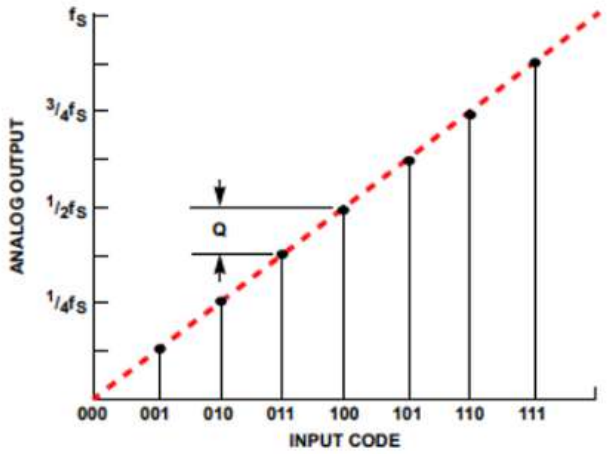


Rebuilding the Analog World: DAC Fundamentals

The Inverse Operation: A DAC converts a digital code (N-bit binary number) into a proportional analog output (typically voltage or current).

Core Architectures: The primary challenge in DAC design is accurately creating the binary-weighted contributions for each bit.

- Binary Weighted Resistor DAC: Conceptually simple, uses resistors with values R, 2R, 4R...
- R-2R Ladder DAC: A more sophisticated approach using only two resistor values (R and 2R), making it far more suitable for IC fabrication.
- String DAC: Uses a series string of 2^N identical resistors and switches.



Key DAC Specifications: The same static error terms apply (DNL, INL, Offset, Gain), but are defined by the output deviation rather than input transition points. Monotonicity is a critical DAC specification.

Replay (Ctrl+Alt+R)

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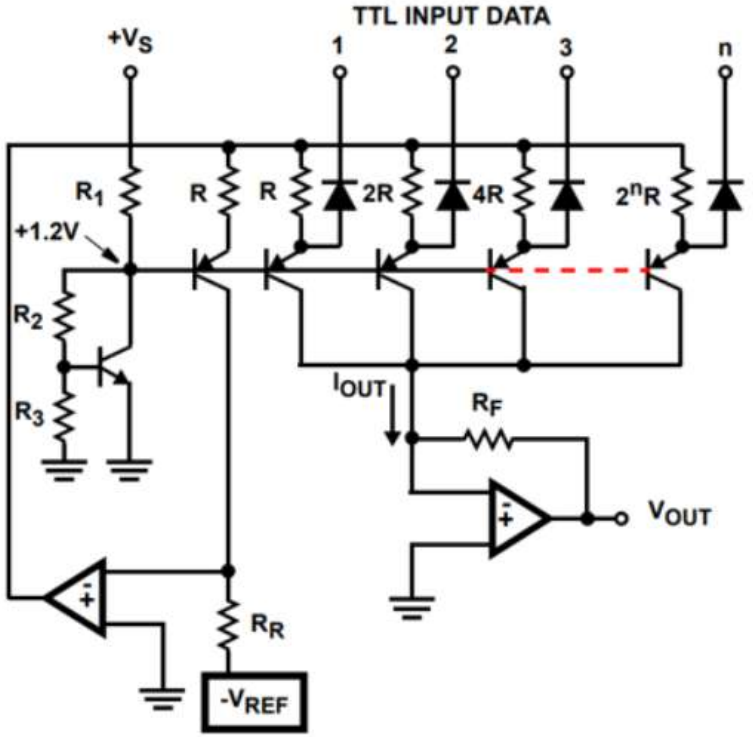
Binary Weighted Resistor DAC

Major Limitation: Resistor Range

$$V_{OUT} = -R_f \cdot V_{REF} \cdot \left(\frac{b3}{R} + \frac{b2}{2R} + \frac{b1}{4R} + \frac{b0}{8R}\right)$$

- For an N-bit DAC, the resistor values must span a range of 2^{N-1} .
- For a 12-bit DAC, this is a 2048:1 ratio (e.g., 1 kΩ to 2.048 MΩ).

- Fabricating such a wide range of resistors with the required precision and ensuring they track over temperature is extremely difficult and costly on a monolithic IC. This makes the architecture impractical for high-resolution DACs.

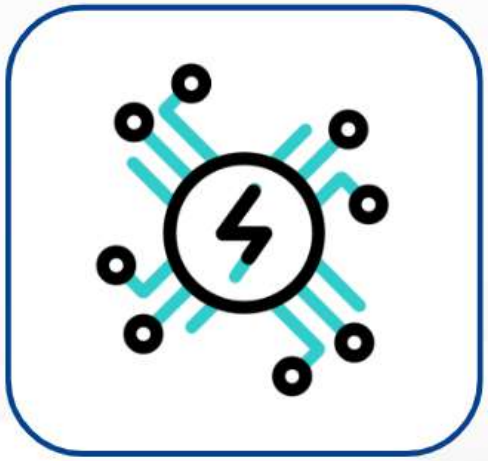


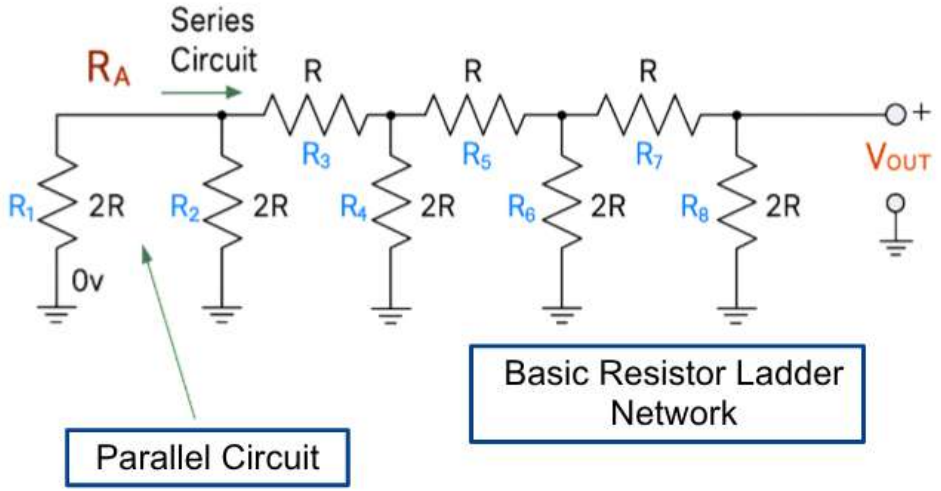
R-2R DAC: Circuit Analysis

Analysis via Thevenin's Theorem and Superposition.

Equivalent Resistance:

- By repeatedly applying Thevenin's theorem starting from the LSB, it can be shown that the resistance looking back into any node towards the LSB side is always R. The total equivalent resistance of the ladder is R.





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R-2R DAC: Circuit Analysis

The contribution to the output voltage from the MSB is:

$$V_{OUT,MSB} = -V_{REF} \cdot \frac{R_f}{2R}$$

Similarly, the contribution from the next bit (b N−2) is half of the MSB's contribution:

$$V\ OUT, MSB - 1\ = -V\ REF\ x\ Rf\ /4R\ .$$

General Output Equation: By superposition, the total output is the sum of the contributions from each bit.

$$V_{OUT} = -V_{REF} \cdot \frac{R_f}{R} \cdot \sum_{i=0}^{N-1} \frac{b_i}{2^{N-i}}$$

$$V_{OUT} = -V_{REF} \cdot \left(\frac{\text{Digital Code}}{2^N} \right)$$

